

CS 3305 - Sample Runs for Assignment 9 Graphs

===== Test 1 =====

Input Matrix:

```
1 1
1 1
```

Reachability Matrix:

```
3 3
3 3
```

In-degrees:

Node 1 in-degree is 2
Node 2 in-degree is 2

Out-degrees:

Node 1 out-degree is 2
Node 2 out-degree is 2

Total number of self-loops: 2

Total number of cycles of length 2 edges: 4

Total number of paths of length 1 edge: 4

Total number of paths of length 2 edges: 8

Total number of paths of length 1 to 2 edges: 12

Total number of cycles of length 1 to 2 edges: 6

===== Test 2 =====

Input Matrix:

```
1 1 1
1 1 1
1 1 1
```

Reachability Matrix:

```
13 13 13
13 13 13
13 13 13
```

In-degrees:

Node 1 in-degree is 3
Node 2 in-degree is 3
Node 3 in-degree is 3

Out-degrees:

Node 1 out-degree is 3
Node 2 out-degree is 3
Node 3 out-degree is 3

Total number of self-loops: 3

Total number of cycles of length 3 edges: 27

Total number of paths of length 1 edge: 9

Total number of paths of length 3 edges: 81

Total number of paths of length 1 to 3 edges: 117

Total number of cycles of length 1 to 3 edges: 39

===== Test 3 =====

Input Matrix:

```
1 1 1 1
1 1 1 1
1 1 1 1
1 1 1 1
```

Reachability Matrix:

85	85	85	85
85	85	85	85
85	85	85	85
85	85	85	85

In-degrees:

Node 1 in-degree is 4
Node 2 in-degree is 4
Node 3 in-degree is 4
Node 4 in-degree is 4

Out-degrees:

Node 1 out-degree is 4
Node 2 out-degree is 4
Node 3 out-degree is 4
Node 4 out-degree is 4

Total number of self-loops: 4

Total number of cycles of length 4 edges: 256

Total number of paths of length 1 edge: 16

Total number of paths of length 4 edges: 1024

Total number of paths of length 1 to 4 edges: 1360

Total number of cycles of length 1 to 4 edges: 340

===== Test 4 =====

Input Matrix:

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Reachability Matrix:

781	781	781	781	781
781	781	781	781	781
781	781	781	781	781
781	781	781	781	781
781	781	781	781	781

In-degrees:

Node 1 in-degree is 5
Node 2 in-degree is 5
Node 3 in-degree is 5
Node 4 in-degree is 5
Node 5 in-degree is 5

Out-degrees:

Node 1 out-degree is 5
Node 2 out-degree is 5
Node 3 out-degree is 5
Node 4 out-degree is 5
Node 5 out-degree is 5

Total number of self-loops: 5

Total number of cycles of length 5 edges: 3125

Total number of paths of length 1 edge: 25

Total number of paths of length 5 edges: 15625

Total number of paths of length 1 to 5 edges: 19525

Total number of cycles of length 1 to 5 edges: 3905